Balance and Swap: practice at home

Name _____ Date _____

About 15 minutes, a few days after the lesson. Use two plates and small things, or draw both sides. A grown-up may read the questions aloud. Wrong answers are fine: they show what to try next.

No.1

Start here

Fill in the missing number: 6 + 3 = 5 + ▢

No.2

Start here

Fill in the missing number: 10 + 4 = 9 + ▢

No.3

Start here

Fill in the missing number: 20 + 30 = 40 + ▢

No.4

Think harder

Ravi says 36 + 14 and 34 + 16 have the same total. Do not add. Is he right?

Circle Yes or No. Then write: I know because _____.

Yes No

No.5

Go further

Put 12 small things in a pile of 7 and a pile of 5. Give 1 across and write the new sum, like 6 + 6 = 12. Write all the sums you can find. Many answers are right.

Answers for grown-ups

Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. **4**
6 + 3 is 9, and 5 + 4 is 9. The 6 became 5 (1 less), so the 3 becomes 4 (1 more).

2. **5**
10 went down by 1 to 9, so 4 goes up by 1 to 5. Both sides make 14.

3. **10**
20 went up by 20 to 40, so 30 goes down by 20 to 10. Both sides make 50.

4. **Yes. 36 went down by 2 to 34 and 14 went up by 2 to 16. Both make 50.**
One number goes down by 2 and the other goes up by 2, so the total stays the same.

5. **Any list that shows the first number going down by 1 and the second going up by 1, for example 7 + 5, 6 + 6, 5 + 7, 4 + 8, 3 + 9, 2 + 10, 1 + 11, 0 + 12.**
Each time one thing moves across, the total stays 12. A grown-up can count the things on both piles to check.

More practice online (no timers, no scores): ask your tutor for the practice link.